DE–2304

Sub. Code
13

DISTANCE EDUCATION

B.Sc.(Mathematics) DEGREE EXAMINATION, DECEMBER 2015.

CLASSICAL ALGEBRA

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) (i) Prove that every convergent sequence is a bounded sequence.

(ii) If $(an) \to a$, $(bn) \to b$ and $an \le bn$ for all n then prove that $a \le b$.

(b) Let $a_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}$. Show that (a_n) diverges to infinity.

2. (a) Test the convergence of

(i) $\sum_{n=1}^{\infty} \frac{3^n n!}{n^n}$

(ii) $\sum_{n=1}^{\infty} \frac{1}{4n^2 - 1}$.

(b) Show that the series $\frac{1}{1^p} + \frac{1}{2^p} + \frac{1}{3^p} + \cdots$ is convergent when $p > 1$, and divergent when $p \le 1$.

3. (a) (i) Find the sum to infinity the series $\frac{1}{6}+\frac{1\cdot 4}{6\cdot 12}+\frac{1\cdot 4\cdot 7}{6\cdot 12\cdot 18}+\cdots$.

(ii) Show that if $|y|<\frac{1}{3}\cdot|x|$, then

$$1+n\left(\frac{2y}{x+y}\right)+\frac{n(n+1)}{1\cdot 2}\left(\frac{2y}{x+y}\right)^2+\frac{n(n+1)(n+2)}{1\cdot 2\cdot 3}\left(\frac{2y}{x+y}\right)^3+\cdots=\left(\frac{x+y}{x-y}\right)^n.$$

(b) Show that $\frac{3}{10}\left[\log 10+\frac{1}{2^7}+\frac{1}{2}\cdot\frac{3}{2^{14}}+\frac{1}{3}\cdot\frac{3^2}{2^{21}}\right]=\log 2$.

4. (a) Show that

$$\log\left(\frac{n+1}{n}\right)=2\left[\frac{1}{2n+1}+\frac{1}{3(2n+1)^3}+\frac{1}{5(2n+1)^5}+\cdots\right].$$

(b) Prove that

$$\log\left(1+\frac{1}{n}\right)^n=1-\frac{1}{2(n+1)}-\frac{1}{2\cdot 3(n+1)^2}-\frac{1}{3\cdot 4(n+1)^3}-\cdots$$

5. (a) Find the condition that the equation $x^4+px^3+qx^2+rx+s=0$ should have two roots α and β connected by the notation $\alpha+\beta=0$.

(b) Find the sum of the fifth powers of the roots of $x^4-7x^2-4x-3=0$.

6. (a) Solve the equation $x^4+20x^3-143x^2+430x+462=0$ by removing its second term.

(b) Solve the equation $x^4-6x^3+12x^2-10x+3=0$ given that it has multiple roots.

7. (a) (i) Show that $1!\ 3!\ 5!\cdots(2n-1)! > (n!)^n$.

(ii) Show that $n^n > 1\cdot 3\cdot 5\cdots(2n-1)$.

(b) Solve, by using matrix methods the simultaneous equations

$x+2y-z=2$; $3x+8y+2z=10$; $4x+9y-z=12$.

8. (a) Solve : $x+y+2z=4$; $2x-y+3z=9$; $3x-y-z=2$ using Cramer's rule.

(b) Find the eigen values and eigen vectors of the matrix $\begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$.

DE–2305

Sub. Code
14

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, DECEMBER 2015.

CALCULUS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Differentiate $\sin(\cos x)$ and $\log(\cos x)$.

   (b) Find the maxima and minima of the function $2(x^2 - y^2) - x^4 + y^4$.

2. (a) Find the equation of the tangent to the curve $y = \dfrac{6x}{x^2 - 1}$ at the point $(2, 4)$.

   (b) What is the radius of curvature of the curve $x^4 + y^4 = 2$ at the point $(1,1)$?

3. (a) Evaluate

   (i) $\displaystyle\int \frac{dx}{\sqrt{a^2 + x^2}}$.

   (ii) $\displaystyle\int \sqrt{a^2 - x^2}\, dx$.

   (b) Evaluate $\displaystyle\int \frac{x + \sin x}{1 + \cos x} dx$

4. (a) Prove that $\int_0^{\pi/4} \log(1+\tan\theta)\,d\theta = \frac{\pi}{8}\log 2$.

(b) Establish a reduction formula for $\int \cos^n x\,dx$ and hence find $\int_0^{\pi/2} \cos^8 x\,dx$.

5. (a) Form the differential equation by eliminating α and β from $(x-\alpha)^2+(y-\beta)^2=r^2$.

(b) Solve $xyP^2+P(3x^2-2y^2)-6xy=0$.

6. (a) Solve $x^2\frac{d^2y}{dx^2}+2x\frac{dy}{dx}=6x^2+2x+1$.

(b) Solve : $(D^2+6D+9)y=e^{-3x}$.

7. (a) Find : $L[t^5+\cos 2t]$.

(b) Find $L^{-1}\left\{\frac{2S}{(S^2-4)^2}\right\}$ as a function of t.

8. (a) Solve $p^2+q^2-2px-2qy+1=0$.

(b) Solve $(y+z-x)p+(z+x-y)q=x+y-z$.

DE–2306

Sub. Code
15

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, DECEMBER 2015.

ANALYTICAL GEOMETRY AND VECTOR CALCULUS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Show that the equation of the pair of straight lines each inclined at an angle 45° to one of other of the lines given by the equation $ax^2 + 2hxy + by^2 = 0$ is $(a - 2h + b)x^2 + 2(a - b)xy + (a + 2h + b)y^2 = 0$.

   (b) Find the equation of the circle described on the line joining the points (1, 2) and (2, 4) as its diameter.

2. (a) Show that the circles $x^2 + y^2 - 6x - 9y + 13 = 0$ and $x^2 + y^2 - 2x - 16y = 0$ touch each other and find the co-ordinates of the point of contact.

   (b) Find the locus of the foot of the perpendiculars drawn from the pole to the tangents to the circle $r = 2a\cos\theta$.

3. (a) Find the equation of the plane which passes through the point (–1,3, 2) and perpendicular to the two planes $x + 2y + 2z = 5$ and $3x + 3y + 2z = 8$.

   (b) Show that the straight lines whose dc's given by $2l - m + 2n = 0$ and $1m + mn + nl = 0$ are at right angles.

4. (a) Find the image of the point (1, –2, 3) in the plane $2x - 3y + 2z + 3 = 0$.

(b) Show that the lines $\frac{x-2}{1} = \frac{y-4}{2} = \frac{z-5}{2}$ and $\frac{x-5}{2} = \frac{y-8}{3} = \frac{z-7}{2}$ are co-planar and find the equation of the plane containing them.

5. (a) Find the shortest distance and the equation of the line of shortest distance between the straight line $\frac{x+3}{-4} = \frac{y-6}{6} = \frac{z}{2}$ and $\frac{x+2}{-4} = \frac{y}{1} = \frac{z-7}{1}$.

(b) Obtain the equation of the sphere through the points (1, 1, 2), (2, –2, 3) and having its centre on the line $2x + 3y = 0 = 5x + y - z$.

6. (a) Find the equation of the sphere whose radius is 4 and centre at (1, 2,3).

(b) Find the equation of the sphere for which the given circle as a great circle $x^2 + y^2 + z^2 + 7y - 2z + 2 = 0$, $2x + 3y + 4z = 8$.

7. (a) If $\nabla \Phi = 2yz^3\bar{i} + x^2z^3\bar{j} + 3x^2yz^2\bar{k}$ find Φ if $\Phi\ (1, -2, 2) = 4$.

(b) Prove that $\text{curl}(r \times a) = -2a$ and $\text{div}(r \times a) = 0$, where r is a positive vector and a is constant vector.

8. (a) Evaluate $\int_C f \cdot dr$, where $f = (x^2 + y^2)i - 2xyj$ and the curve C is the rectangle in the xy plane bounded by $y = 0, y = b;\ x = 0, x = a$.

(b) Verify Stoke's theorem for the vector function $f = y^2\vec{i} + y\vec{j} - xz\vec{k}$ and S is the upper half of the sphere $x^2 + y^2 + z^2 = a^2$ and $z \geq 0$.

DE–2307

Sub. Code
23

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, DECEMBER 2015.

MECHANICS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) State and prove the converse of triangle of forces.

   (b) The resultant of two forces P, Q acting at a certain angle is X and that of P, R acting at the same angle is also X. The resultant of Q, R again acting at the same angle is Y. Prove that $P = \left(X^2 + QR\right)^{1/2} = \dfrac{QR(Q+R)}{Q^2 + R^2 - Y^2}$. Also prove that, if $P + Q + R = 0, Y = X$.

2. (a) Two forces P and Q have a resultant R and the resolved part of R in the direction of P is of magnitude Q. Show that the angle between the forces in $2\sin^{-1}\sqrt{\dfrac{P}{2Q}}$.

   (b) Forces $F_1, F_2, F_3, F_4, F_5, F_6$ act along the sides of a regular hexagon taken in order. Show that they will be in equilibrium if

   (i) $F_1 + F_2 + F_3 + F_4 + F_5 + F_6 = 0$ and

   (ii) $F_1 - F_4 = F_3 - F_6 = F_5 - F_2$.

3. (a) State and prove Varignon's theorem.

   (b) If three coplanar forces acting on a rigid body keep it in equilibrium, show that they must either be concurrent or be all parallel.

4. (a) If a body is at rest on a rough plane inclined to the horizon at an angle greater than the angle of friction and is acted upon by a force, parallel to the plane and along the line of greatest slope, find the limits between which the force must lie.

   (b) Explain the terms, coefficient of friction, angle of friction, cone of friction.

5. (a) A particle is projected with velocity $\sqrt{2ga}$ from a point at a height h above a level plain. Show that the maximum range is $2\sqrt{a(a+h)}$.

   (b) Determine when the horizontal range of a projectile is maximum, given the magnitude u of the velocity of projection.

6. (a) Two equal marble balls A, B lie in a horizontal circular groove at the opposite ends of a diameter, A is projected along the groove and after time t, impinges on B, show that a second impact takes plane after a further interval $\frac{2t}{e}$.

   (b) A smooth sphere of mass m collides obliquely with a fixed smooth plane with a velocity v inclined to the normal to the plane at an angle α. Find its velocity of rebound, the impulse imported to it and the change in its kinetic energy.

7. (a) Discuss the composition of two Simple Harmonic Motions (SHM) of the same period and in the same straight line.

(b) Show that the energy of a system executing S.H.M. is proportional to the square of the amplitude and of the frequency.

8. (a) Find the law of force towards pole under which the curve $r^n = a^n \cos n\theta$ can be described.

(b) A particle moves in a plane with an acceleration which is always directed to a fixed point O in the plane, to obtain the differential equation of its path.

DE–2308

Sub. Code
24

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, DECEMBER 2015.

ANALYSIS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each questions carry equal marks.

(5 × 20 = 100)

1. (a) Prove : If $(M_1, d_1), (M_2, d_2), .., (M_n, d_n)$ are metric spaces then $M_1 \times M_2 \times ... \times M_n$ is a metric space with metric d defined by $d(x, y) = \sum_{i=1}^{n} d_i(x_j, y_i)$ where $x = (x_1, x_2, ..., x_n)$; $y = (y_1, y_2, ..., y_n)$. (10)

   (b) Let (M, d) be a metric space. Define $\rho(x, y) = \dfrac{d(x, y)}{1 + d(x, y)}$ prove that d and ρ are equivalent metrics on M. (10)

2. (a) Prove $(M_1, d_1) \to (M_2, d_2)$ is continuous iff $f^{-1}(G)$ is open in M_1. Whenever G is open in M_2. (10)

   (b) Prove : Let M be a metric space and $A \subseteq M$, then $\overline{A} = A \cup D(A)$. (10)

3. (a) State and prove the Baire's category theorem. (10)

   (b) Prove that a closed set A in a metric space M is nowhere dense iff A^C is everywhere dense. (10)

4. (a) Prove that a subspace of R is connected iff it is an interval. (10)

   (b) Prove : A metric space *(M, d)* is totally bounded iff every sequence in M has a Cauchy subsequence.(10)

5. In metric space M, prove that the following statements are equivalent :

   (a) M is compact.

   (b) Any infinite subset of M has a limit point.

   (c) M is sequentially compact.

   (d) M is totally bounded and complete. (20)

6. (a) State and prove Heine Borel theorem. (10)

   (b) Prove that a metric space M is compact iff any family of closed sets with finite intersection property has non-empty intersection. (10)

7. (a) Prove : Let $f_n : [0,1] \to R$ be defined by $f_n(x) = \frac{x}{n}$. The sequence (f_n) converges pointwise to $f_n : [0,1] \to R$ defined by $f(x) = 0$. (10)

   (b) State and prove Dini's theorem. (10)

8. State and prove Piccard's theorem. (20)

DE–2309

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25

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, DECEMBER 2015.

PROBABILITY AND STATISTICS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) State and prove Baye's theorem.

   (b) Mr. X and Mr. Y throw alternatively with a pair of ordinary dice. Mr. X wins if he throws 6 before Mr. Y throws 7 and Mr. Y wins if he throws 7 before Mr. X throws 6. If Mr. X begins the game show that the chance of his winning is 30/61.

2. (a) Prove that the sum of the squares of the deviations of a set of n values is minimum when the deviations are taken from their mean.

   (b) A random variable X has the following probability distribution.

x_i :	–2	–1	0	1	2	3
$P(x_i)$:	0.1	k	0.2	$2k$	0.3	k

   Find : (i) k (ii) mean (iii) variance (iv) $p(x \geq 2)$ (v) $p(x < 2)$.

3. (a) Fit a straight line to the following data and estimate the value of y corresponding to $x = 6$

x :	0	5	10	15	20	25
y :	12	15	17	22	24	30

(b) 10 students obtained the following percentage of marks in the college internal test X and in the final University examination Y. Find the correlation coefficient between the marks of the 2 tests.

X :	51	63	63	49	50	60	65	63	46	50
Y :	49	72	75	50	48	60	70	48	60	56

4. (a) Prove that the angle between two regression lines is

$$\theta = \tan^{-1}\left[\left(\frac{\gamma^2 - 1}{\gamma}\right)\left(\frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2}\right)\right].$$

(b) 10 competitors in a beauty contest were ranked by 3 judges in the following order.

J_1 :	1	6	5	10	3	2	4	9	7	8
J_2 :	3	5	8	4	7	10	2	1	6	9
J_3 :	6	4	9	8	1	2	3	10	5	7

Use rank correlation coefficient to discuss which pair of judges have the nearest approach to common tests in beauty.

5. (a) Derive a formula to find mean and variance for the binomial distribution.

(b) Fit a Poisson distribution for the following data :

x :	0	1	2	3	4	Total
f :	123	59	14	3	1	200

6. (a) In a distribution exactly normal, 7% of the items are under 35 and 89% are under 63. What are the mean and standard deviation of the distribution?

(b) Derive the formula for moment generating function for Poisson distribution.

7. (a) A group of 10 rats fed on a diet A and another group of 8 rats fed on a different diet B recorded the following increase in weight in gms.

Diet A : 5 6 8 1 12 4 3 9 6 10

Diet B : 2 3 6 8 1 10 2 8

Test whether diet A is superior to diet B.

(b) In one sample of 8 observations the sum of the squares of deviation of the sample values from the sample mean was 84.4 and in another sample of 10 observations it was 102.6. Test whether the difference in variance is significant at 5% level using F -test.

8. (a) Construct with the help of the data given below, Fisher's Index Number and show that it satisfies both the factor reversal test and time reversal test.

Commodity	A	B	C	D
Base year price in Rs. :	5	6	4	3
Base year Quantity in kg :	50	40	120	30
Current year price in Rs. :	7	8	5	4
Current year quantity in kg. :	60	50	110	35

(b) Construct the cost of living index number in 2005.

Group :	Food	Rent	Clothing	Fuel	Miscellaneous
% of expense :	40	15	15	10	20
Price in 2000 :	150	50	100	20	60
Price in 2005 :	174	60	125	25	90

DE–9160

Sub. Code
31

DISTANCE EDUCATION

B.Sc. DEGREE EXAMINATION, DECEMBER 2015.

Mathematics

ALGEBRA

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) If A, B and C are any three finite sets prove that $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C|$. (6)

(b) Let ρ be an equivalence relation defined on a set S. Then prove that

(i) $a \rho b \Leftrightarrow [a] = [b]$

(ii) Any two distinct equivalence classes are disjoint

(iii) S is the union of all the equivalence classes. (8)

(c) $f : A \to B$, $g : B \to C$ be bijections. Then prove that $g \circ f : A \to C$ is also a bijection. (6)

2. (a) If ρ and σ are equivalence relations defined on a set S, prove that $\rho \wedge \sigma$ is also an equivalence relation. (6)

(b) Show that $f : R \to (0,1)$ defined by $f(x) = \frac{1}{2}\left[1 + \frac{x}{1+|x|}\right]$ is a bijection. (6)

(c) Prove : If A, B and C are any three sets then

(i) $(A \cup B) \times C = (A \times C) \cup (B \times C)$

(ii) $(A - B) \times C = (A \times C) - (B \times C)$. (8)

3. (a) Let G be the set of all real number except -1 define * on G by $a*b = a+b+ab$. Then prove that $(G,*)$ is a group. (10)

(b) Prove : Let A and B be two subgroups of a group G, then AB is a subgroup of G iff $AB = BA$. (10)

4. (a) State and prove lagrange's theorem on group. (6)

(b) Prove that isomorphism is an equivalence relation among groups. (8)

(c) H_a is the set of all elements in G which commute with 'a'. Show that H_a is a subgroup of G. (6)

5. (a) Prove that the ring of Gaussian integers $R = \{a+bi / a,b \in z\}$ is an Eucledian domain, where $d(a+bi) = a^2 + b^2$. (6)

(b) In an integral domain D of characteristic p, prove that the order of every element in the additive group is p. (8)

(c) A non-empty subset S of a ring R is a subring iff $a,b \in S \Rightarrow a-b \in S$ and $ab \in S$. (6)

6. (a) Prove that the set F of all real numbers of the form $a+b\sqrt{2}$, where $a,b \in Q$ is a field under the usual addition and multiplication of real numbers. (10)

(b) Prove : The field of quotients F of an integral domain D is the smallest field containing D$(i.e)$. If F' is any other field containing D then F' contains a subfield isomorphic to F. (10)

7. (a) Let V be a finite dimensional inner product space. Let W be a subspace of V. Then prove that V is the direct sum of W and $W^{\perp}$. (10)

(b) Let V be a vector space over a field F and S be a non-empty subset of V, then prove that

(i) $L(S)$ is a subspace of V.

(ii) $S \subseteq L(S)$

(iii) If W is any subspace of V such that $S \subseteq W$ then $L(S) \subseteq W$. (10)

8. (a) Let V and W be vector spaces over a field F and $T : V \to W$ be an epimorphism, then prove that

(i) $\ker(T) = V_1$ is a subspace of V and

(ii) $V / V_1 = W$. (10)

(b) Prove that every finite dimensional inner product space has orthonormal basis. (10)

DE–2311

Sub. Code
32

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, DECEMBER 2015.

OPERATIONS RESEARCH

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

$(5 \times 20 = 100)$

1. (a) Solve graphically,

Maximize : $Z = 5x_1 + 3x_2$

Subject to the constraints : $3x_1 + 5x_2 \leq 15$

$5x_1 + 2x_2 \leq 10$

$x_1, x_2 \geq 0.$ (10)

(b) Discuss the various phases of operations research. (10)

2. (a) Solve the following LPP, by using duality principle :

Minimize : $Z = 2x_1 + 2x_2$

Subject to : $2x_1 + 4x_2 \geq 1$

$x_1 + 2x_2 \geq 1$

$2x_1 + x_2 \geq 1$ and

$x_1, x_2 \geq 0.$ (10)

(b) Use simplex method to solve the following LPP :

Maximize : $Z = 10 + x_1 + x_2 + 2x_3$

Subject to : $14x_1 + x_2 - 6x_3 + 3x_4 = 7$

$$16x_1 + \frac{1}{2}x_2 - 6x_3 \le 5$$

$$3x_1 - x_2 - x_3 \le 0$$

$$x_1, x_2, x_3, x_4 \ge 0.$$ (10)

3. (a) Write the algorithm for Big-M method. (10)

(b) Solve the following LPP using Two-phase method :

Minimize : $Z = 60x_1 + 80x_2$

Subject to : $20x_1 + 30x_2 \ge 900$

$$40x_1 + 30x_2 \ge 1200$$

$$x_1, x_2 \ge 0.$$ (10)

4. (a) Find the optimum solution of the following integer programming problem :

Maximize : $Z = 2x_1 + 3x_2$

Subject to : $2x_1 + 2x_2 \le 7$

$x_1 \le 2;\ x_2 \le 2;\ x_1, x_2 \ge 0$ and are integers.

(10)

(b) Explain the cutting plane algorithm. (10)

5. (a) Solve the following Transportation problem by MOD 1 method : (10)

$$\begin{array}{c|ccc|c} & X & Y & Z & \\ \hline A & 8 & 7 & 3 & 60 \\ B & 3 & 8 & 9 & 70 \\ C & 11 & 3 & 5 & 80 \\ \hline & 50 & 80 & 80 & \end{array}$$

(b) Find the initial basic feasible solution to the below transportation problem by North West corner rule

11	13	17	14	250
16	18	14	10	300
21	24	13	10	400
200	225	274	250	

(10)

6. (a) Solve the following assignment problem : (10)

	1	2	3	4
A	10	12	19	11
B	5	10	7	8
C	12	14	13	11
D	8	15	11	9

(b) Obtain the initial basic feasible solution to the following transportation problem by least cost method : (10)

To

From

2	7	4	5
3	3	1	8
5	4	7	7
1	6	2	14
7	9	18	34

7. (a) Solve the following game graphically : (10)

Player *B*

Player *A*

1	–3
3	5
–1	6
4	1
2	2
–5	0

(b) Consider the game G with the following pay off :

$$\begin{array}{cc} & \text{Player } B \\ \text{Player A} & \begin{pmatrix} 2 & 6 \\ -2 & \mu \end{pmatrix} \end{array}$$

(i) Show that G is strictly determinable whatever μ may be.

(ii) Determine the value of G. (10)

8. (a) Show how to solve a 2×2 two person zero sum game without any saddle point. Derive the expression for optimal strategies and the value of game. (10)

(b) Discuss about PERT algorithm. (10)

DE–2312

Sub. Code
33

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, DECEMBER 2015.

NUMERICAL METHODS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Find the positive root of $x-\cos x=0$ by bisection method.

   (b) Find the positive root of $x^3=2x+5$ by false position method.

2. (a) Solve by the Gauss-Jordan method the following system of equations.

   $10x+y+z=12$; $2x+10y+2=13$; $x+y+5z=7$.

   (b) Solve the following system of equations by triangularization method

   $x+y+z=1$; $4x+3y-z=6$; $3x+5y+3z=4$.

3. (a) Evaluate :

(i) $\Delta^3(1-x)(1-2x)(1-3x)$ when $h=1$

(ii) $\Delta^{10}(1-x)(1-2x)(1-3x)\cdots(1-10x)$ taking $h=2$

(iii) $\Delta^{10}(1-x)(1-2x)(1-3x)\cdots(1-10x)$ taking $h=1$

(b) State and prove Newton's forward interpolation formula.

4. (a) Find the missing term in the following data set :

x :	1	2	3	4	5	6	7
y :	2	4	8	–	32	64	128

(b) Using Everett's formula, estimate $y(30)$, given :

x :	20	28	36	44
$y(x)$:	2854	3162	7088	7984

5. (a) Show that the nth divided difference of a polynomial of degree n is constants.

(b) From the following table of values of x and y, find $\frac{dy}{dx}$ at $x=1.10$.

x :	1.00	1.05	1.10	1.15	1.20	1.25	1.30
y :	1.00000	1.02470	1.04881	1.07238	1.09544	1.11803	1.14017

6. (a) Evaluate $\int_0^1 \frac{dx}{1+x}$ using Simpson's $\frac{3}{8}$ rule taking $h=\frac{1}{6}$.

(b) Evaluate the integral $I=\int_4^{5.2} \log_e x\, dx$ using Weddle's rules.

7. (a) Using Taylor's method, find $y(0.1)$ correct to three decimal places from $\frac{dy}{dx} = 1 - 2xy$, $y_0 = 0$.

(b) Solve $y' = 2x - y$, $y(1) = 3$ by Picard's method.

8. (a) Using Euler's method, solve $y' = x + y$, $y(0) = 1$ for $x = 0.0(0.2)(1.0)$.

(b) Use Runge-Kutta method of fourth order to find $y(0.1)$ given that $\frac{dy}{dx} = \frac{1}{x+y}$, $y(0) = 1$.

DE–2313

Sub. Code
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DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, DECEMBER 2015.

COMPLEX ANALYSIS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Prove that the function $f(z)=\begin{cases} \dfrac{xy^2}{x^2+y^2} & \text{if } z\neq 0 \\ 0 & \text{if } z=0 \end{cases}$ satisfies C-R equations at $z=0$ but the function is not differentiable at $z=0$.

   (b) If $f(z)$ is analytic, prove that $\left(\dfrac{\partial^2}{\partial x^2}+\dfrac{\partial^2}{\partial y^2}\right)\left|f(z)^2\right|=4\left|f'(z)\right|^2$.

2. (a) Explain stereographic projection.

   (b) Find the radius of convergence of the following power series $\sum_{n=1}^{\infty}\dfrac{1}{n^p}z^n$.

3. (a) Show that the transformation $w=z^2$ transforms the families of lines $x=h$ and $y=k$ into confocal parabolas, having $w=0$ as the common focus.

   (b) Define cross ratio. Prove that any bilinear transformation having two finite fixed points α and β can be written in the form $\dfrac{w-\alpha}{w-\beta}=k\left(\dfrac{z-\alpha}{z-\beta}\right)$.

4. (a) Find the image of the strip $2<x<3$ under $w=\frac{1}{z}$.

(b) Prove that any bilinear transformation which maps the real axis onto unit circle $|\omega|=1$ can be written in the form $\omega=e^{i\lambda}\left(\frac{z-\alpha}{z-\alpha}\right)$. where λ is real. Also prove that this transformation maps the upper half plane $\operatorname{Im} z\geq 0$ onto the unit circular disc $|\omega|\leq 1$ iff $\operatorname{Im}\alpha\geq 0$

5. (a) Prove $\int_C z^{-2}dz=\begin{cases} 0 & \text{if} \quad C \text{ is the unit circle } |z|=1 \\ 4\pi i & \text{if} \quad C \text{ is the unit circle } |z-1|=1 \end{cases}$

(b) Evaluate $\int_C \frac{e^z}{(z+2)(z+1)^2}dz$ where C is $|z|=3$.

6. (a) If f is analytic inside and on a simple closed curve C and if Z is any point inside C, then prove that $f'(z)=\frac{1}{2\pi i}\int_C \frac{f(\zeta)}{(\zeta-z)^2}d\zeta$. (12)

(b) State and prove Liouville's theorem. (8)

7. (a) Expand $\frac{-1}{(z-1)(z-2)}$ as a power series in z in the regions.

(i) $|z|<1$

(ii) $1<|z|<2$ and

(iii) $|z|>2$

(b) Find the residue of $\frac{e^z}{z^2(z^2+9)}$ at its pole.

8. (a) Prove that $\int_C \tan z \, dz = -4\pi i$, where C is a circle $|z| = 2$. (8)

(b) Evaluate $\int_0^{2\pi} \frac{d\theta}{13 + 5\sin\theta}$. (12)

DE–2314

Sub. Code
35

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, DECEMBER 2015.

DISCRETE MATHEMATICS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Construct the truth table for the formula $(\neg P \vee Q) \wedge (\neg Q \vee P)$. Check whether it is a tautology.

   (b) Write down the truth table for the following compound statements and state which of them are tautologies.

   (i) $(Q \vee R) \rightarrow (P \vee \neg R)$

   (ii) $(P / (P \leftrightarrow Q)) \rightarrow Q$

   (iii) $(P \rightarrow Q) \leftrightarrow (\neg P \vee Q)$.

2. (a) Define equivalence of formulae. Show that $(P \rightarrow Q) \wedge (R \rightarrow Q)$ and $(P \vee R) \rightarrow Q$ are equivalent formulae.

   (b) Find a disjunctive, normal form of $(p \wedge \neg (q \vee r)) \vee (((p \wedge q) \vee \neg r) p)$.

3. (a) Define an open statement with an example. Identify pronouns, objects and universe.

   (b) Prove by indirect method that $(\neg q)$, $p \rightarrow q$, $p \vee t \Rightarrow t$.

4. (a) Write down the rules for well-formed formula of predicate calculus.

   (b) Show that :

   (i) $(\forall x)(P(x)) \rightarrow (\exists x)(p(x))$

   (ii) $(\forall x)(P(x) \wedge Q(x)) \leftrightarrow ((\forall x))(P(x) \wedge (\forall x) Q(x))$ are logically valid statements.

5. (a) Define a complete graph. Give examples for complete graphs of two, three, four, and five vertices.

   (b) Prove that a simple graph with n vertices must be connected if it has more than $\frac{(n-1)\ (n-2)}{2}$ edges. Also give an example for a graph with $\frac{(n-1)\ (n-2)}{2}$ edges which is not connected.

6. (a) State and prove Cayley's formula.

   (b) Prove that a graph G without loops is a tree if and only if any two distinct vertices are connected by a unique path.

7. (a) Prove that every cycle has an even number of edges in common with any cut set.

   (b) Prove : The value of any flow in a network N is less than or equal to the capacity of any cut in the network.

8. (a) Write Fulkerson's Algorithm.

   (b) State and prove the max-flow min-cut theorem. Also write down the algorithm to find a maximum flow.